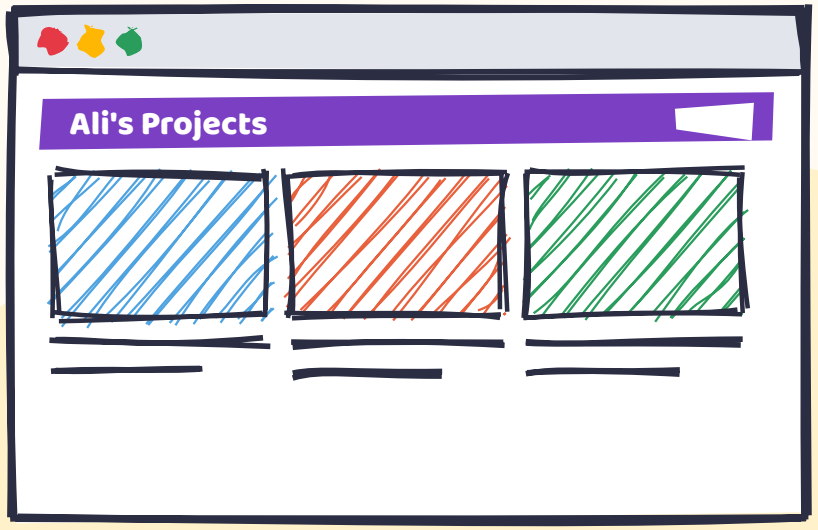
Young Innovators · Class 14

Personal Portfolio Website

Show the world what you have built!

HTML + CSS | JavaScript | Writing about your work



Did you know?
Many engineers get noticed because of the projects they show online, not just their grades.

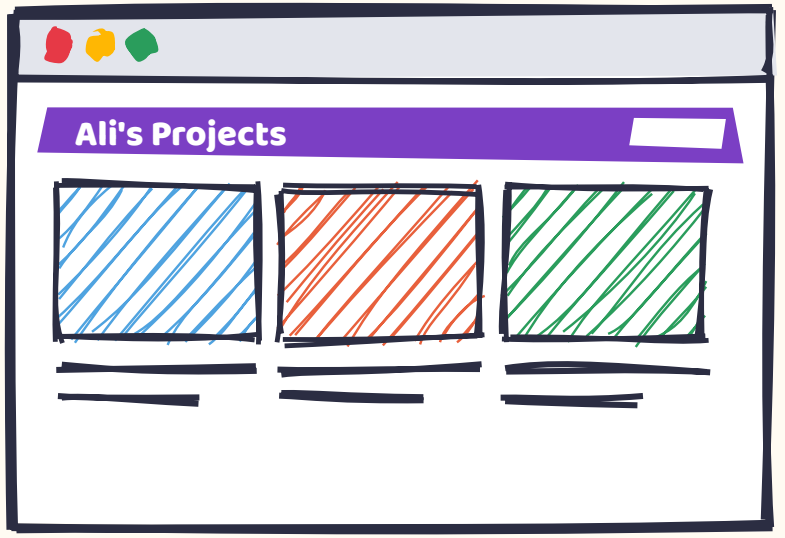
LEARN

Let's think...

Where is all your amazing work?

Lost in your phone

This term you built an alarm, a robot, a plant waterer... The photos are scattered in phones and the code is on school laptops.
One day a **university** or a **competition** asks: **show us what you can do**.



Send 50 photos?

Explain it from memory?

Send ONE link!

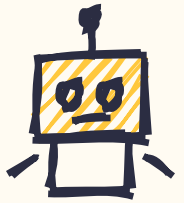
Today: you build your own **portfolio website**: one page per project, with photos and clear explanations.

Learning aims

Today I will learn to…

1. Plan a site with a **sitemap** and a **wireframe**
2. Structure pages with **semantic HTML**
3. Lay out and style with **CSS flexbox and grid**
4. Add **JavaScript**: a dark-mode button or photo gallery
5. Explain a project clearly and **stay safe online**

Keywords
portfolio · sitemap · wireframe · semantic · flexbox · grid · JavaScript



Watch and think

Video time!



HTML, CSS, JavaScript Explained in 4 Minutes

Danielle Thé · about 4 minutes · youtu.be/gT0Lh1eYk78

Watch for...

1. What does **HTML** do? (structure)
2. What does **CSS** do? (style)
3. What does **JavaScript** do? (behaviour)

Scan to watch

After the video: Name one part of your portfolio for each: HTML, CSS and JavaScript.

LEARN

Big idea

A portfolio is a story of your growth



Best work

Choose your 3 to 5 strongest projects, not everything



The process

Sketches, wiring photos, failures and fixes



Reflection

What you learnt and what you would do differently

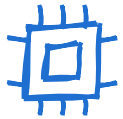
? Which project from this term are you **most proud of**? Which one taught you the most? Are they the same?

Hidden truth

Showing a failure you fixed often impresses people MORE than a perfect project.

Real world

Who uses portfolios?



Engineers
Share code and project write-ups online



Designers
Show app and product designs



University
Applications often ask for projects and activities



Competitions
Judges want to see your process, not just the result

? If a stranger looked at your portfolio for 30 seconds, what should they remember about you?

The 30-second test
Most visitors only skim. Put your best project FIRST.

LEARN

Structure

Semantic HTML: tags with meaning

```
<header>
  <h1>Ali's Engineering Projects</h1>
  <nav>
    <a href="index.html">Home</a>
    <a href="projects.html">Projects</a>
    <a href="about.html">About</a>
  </nav>
</header>
<main>
  <section id="projects"> ... </section>
</main>
<footer>Made by Ali, 2026</footer>
```

Tag	Meaning
<header>	top of the page
<nav>	menu links
<main>	the main content
<section>	a group of content
<footer>	bottom of the page

Why not just div?
Screen readers and search engines understand these tags. Your site is easier to use and to find.

Links between pages

<a href='about.html'> opens another page of your site.

LEARN

Style it

Layout with flexbox and grid

```
nav {
  display: flex;
  gap: 20px;
}
#projects {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(250px, 1fr));
  gap: 16px;
}
.card {
  border-radius: 12px;
  padding: 12px;
  box-shadow: 0 2px 8px rgba(0,0,0,0.15);
}
```

Flexbox

Puts items in a **row** (like menu links) with even gaps.

grid

Makes **columns** of cards. **auto-fill** means: as many columns as fit. On a phone you get 1, on a laptop 3 or 4!

Phones too!
Add <meta name='viewport' content='width=device-width'> in the head so it fits small screens.

LEARN

JavaScript

Add a dark-mode button

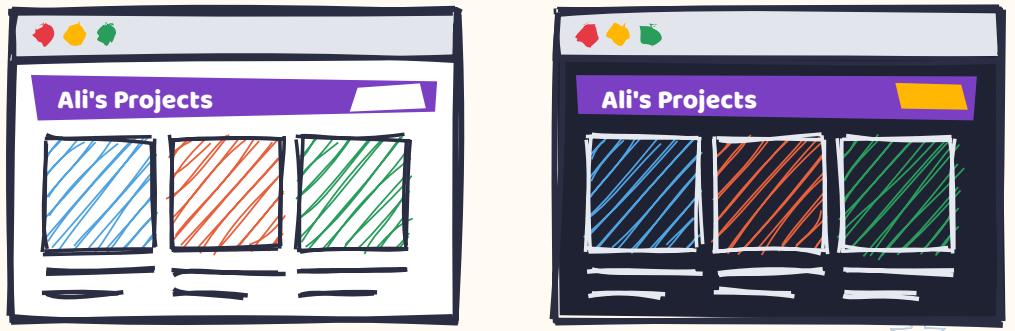
```
<button id="themeBtn">Dark mode</button>

<script>
const btn = document.getElementById("themeBtn");
btn.addEventListener("click", function () {
  document.body.classList.toggle("dark");
});
</script>

/* in CSS */
body.dark {
  --text: #E6E9F0;
  --bg: #1E2233;
}
```

How it works

When the button is **clicked**, JavaScript adds (or removes) the class **dark** on the body. CSS then swaps the colour variables.



toggle = switch on if off, switch off if on. Same idea as the alarm "flip" in Junior Makers!

Communicate

Explain a project in 3 sentences

1. What I built	*“I built an obstacle-avoiding robot car that scans with a sonar sensor on a servo.”*
2. How it works	*“It measures distance left, ahead and right, then turns towards the clearest path.”*
3. What I learnt	*“Sensor readings are noisy, so I averaged 3 readings, which stopped it crashing.”*

Show numbers
“Worked 9 out of 10 times” is stronger than “worked well”.

Be honest
Say what YOU did and what your team did.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Share your work, not your personal life
2. Sitemap and wireframe
3. A project card
4. Pick a colour palette
5. A click-to-enlarge photo gallery
6. Is your site ready?
7. Gallery walk

Stay safe online

Share your work, not your personal life

First name only: no surname, home address, phone number or school timetable.

Photos: show your projects, not faces of other students (unless they and the school agree).

Public by default: pens on CodePen can be seen by anyone. Check settings with your teacher.

No contact form: let the school share your link, instead of strangers messaging you.

Ask first: get permission from a parent and teacher before publishing.



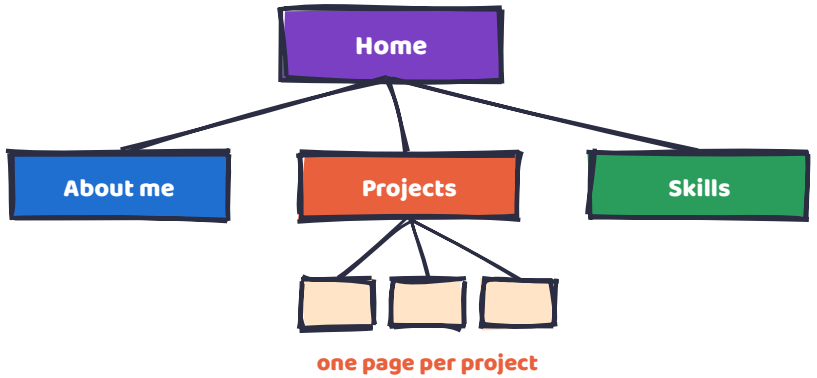
Remember

Once something is online, you may never fully delete it.

Plan first

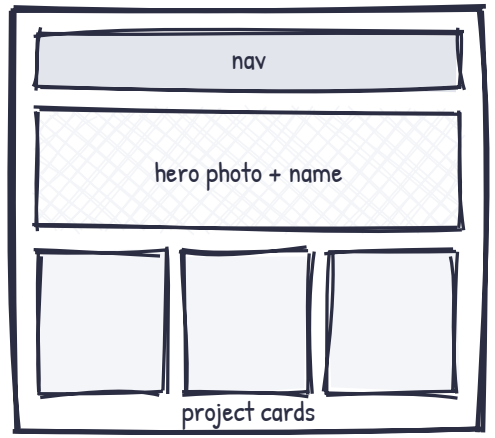
Sitemap and wireframe

Sitemap: which pages?



one page per project

Wireframe: what goes where?



On paper first!

Sketch with a pencil in 10 minutes. Changing a drawing is much faster than changing code.

Notice

No Contact page with your phone or address. More on safety later!

MAKE

Build it

A project card

```
<article class="card">
  <img src="alarm.jpg"
       alt="Breadboard with a PIR sensor, buttons and LEDs">
  <h3>Burglar Alarm with Secret Code</h3>
  <p>A motion alarm that only switches off
     with a 3-button code.</p>
  <a href="alarm.html">Read more</a>
</article>
```



Good alt text
Describe what is IN the photo, not just "photo of project".

Copy, paste, change

Make one card, then copy it for each project and change the picture, title and text.

Photo size
Resize photos to about 1000 px wide before uploading. Big photos make pages slow.

MAKE

Your style

Pick a colour palette

```
:root {
  --main: #7B3FC4;
  --accent: #FFB703;
  --text: #2B2D42;
  --bg: #FFFDF7;
}
body {
  color: var(--text);
  background: var(--bg);
  font-family: Arial, sans-serif;
}
h1, h3 { color: var(--main); }
```

Our palette

- **main** #7B3FC4
- **accent** #FFB703
- **text** #2B2D42
- **background** #FFFDF7

CSS variables

Name a colour **once** (--main) and use it everywhere. Change one line to restyle the whole site!

Design rules

1 main colour, 1 accent, dark text on a light background. Check that text is easy to read.

MAKE

JavaScript: option 2

A click-to-enlarge photo gallery

```
<img id="big" src="robot1.jpg" alt="Robot car, front view">
<img class="thumb" src="robot1.jpg" alt="Robot car, front view">
<img class="thumb" src="robot2.jpg" alt="Robot car wiring">

<script>
const big = document.getElementById("big");
document.querySelectorAll(".thumb").forEach(function (t) {
  t.addEventListener("click", function () {
    big.src = t.src;
    big.alt = t.alt;
  });
});
</script>
```

How it works

Find every **thumb** picture. When one is clicked, copy its **src** and **alt** into the big picture.

Choose ONE

Dark mode OR gallery is enough. Finish one well before starting another.

MAKE

Test like an engineer

Is your site ready?

Check	Done?
Every image has helpful alt text	yes / no
Text is easy to read (contrast)	yes / no
Works on a phone-size window	yes / no
Every link opens the right page	yes / no
No personal information	yes / no
Spelling checked by a friend	yes / no

Debug checklist

Broken image? Check the file name and capital letters exactly.
Layout broken? Look for a missing closing tag or }.

Press F12 in the browser and click the phone icon to see your site at phone size.

Class game

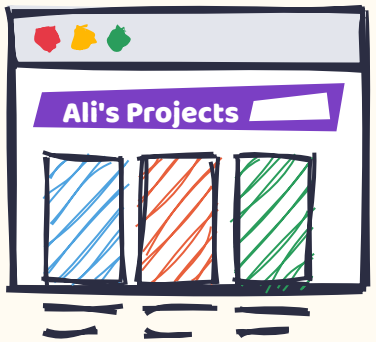
Gallery walk

How to play

1. Open your site on your laptop and leave it on the desk.
2. Everyone walks around with sticky notes.
3. On each site, leave **two stars** (things that are great) and **one wish** (one thing to improve).
4. Go back, read your notes, and fix your top wish.

Then discuss

Was any feedback surprising? What did other people notice that you missed?



Big idea

Real designers test with real users. Fresh eyes find problems you cannot see.

MAKE

Level up!

Portfolio challenges

Choose your challenge



Remember me: save the dark-mode choice with localStorage so it stays after refresh.



Timeline: a page showing your projects in date order with what you learnt each time.



Video: add a short demo clip of a working project (no faces).



Publish: with permission, host it on the school site or a free hosting service.

Think like an engineer

In 5 years, what do you hope your portfolio will show? Which skills do you want to add?

Boss level

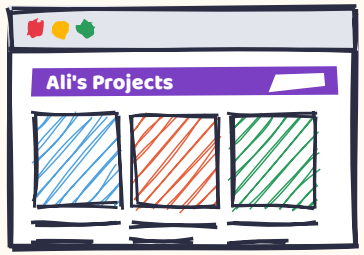
Make the nav menu collapse into a ☰ button on small screens.

Show what you know

What did we learn?

- [x] A portfolio tells the story of your best work and growth
- [x] Plan with a sitemap and a wireframe before coding
- [x] Semantic HTML + CSS flexbox and grid build the layout
- [x] JavaScript reacts to clicks: dark mode or a gallery
- [x] Share your projects, never your personal details

At home
Show your site to a family member. Can they explain one of your projects back to you?



Exit ticket: write your 3-sentence explanation for one project.

Next class:
School Helpdesk Chatbot!